

Saurabh singh ☆☆☆☆☆
H-30, Madhuban CHS, Sector 22, koperkhairane, Navi Mumbai,
talk_pks@yahoo.co.in,
02227541862,

Report of Evidence of Use

The views presented here are of the author only and do not necessarily reflect the views of others. The author is an independent contractor that acts on behalf of neither the patent owner nor any other third party, including ClaimCharts.info. We would like to thank ClaimCharts.info for introducing to us the opportunity to prepare this report. For comments and questions, please feel free to contact us directly, or via my profile at ClaimCharts.info.

Patent Number: USUS8902225B2

Claim Number: (1

What is claimed is: 1. A user interface method, the method comprising: obtaining a projector display of an initial image; using one or more sensors coupled to a computer processor, sensing a user interaction with the displayed initial image, the user interaction being in a three dimensional coordinate system, and said sensing being based on quadrilateral angle navigation that determines a sensing point position within the three dimensional coordinate system, wherein an original image is displayed from a certain point in space preconfigured to match specific points in the quadrilateral angle navigation, and the certain point in space is matched to space location of the user interaction to determine the sensing point, said sensing resulting in each sensor providing coordinate information to the computer processor enabling the computer processor to correlate with the original image in the three dimensional coordinate system; in the computer processor, correlating the sensed user interaction with the three dimensional coordinate system and responsively manipulating the initial image resulting in an updated image based on the correlated user interaction; and from the computer processor through a projector, projecting the resulting updated image, the updated image being equivalent to the initial image manipulated by a distortion in three dimensional space.

Element 1:

What is claimed is: 1. A user interface method, the method comprising:

Tsao discloses a user interface method (For e.g., there is provision in this prior art for barrier-free interaction with 3D images).

See e.g., **Abstract**("This invention relates generally to new and improved methods and apparatus for displaying volumetric 3D images with colors and for barrier-free interaction with the 3D images.").

Element 2:

obtaining a projector display of an initial image;

Tsao discloses obtaining a projector display of an initial image (For e.g., images are projected using volumetric image projector.).

*See e.g.,***2:62-67**("This invention also improves the interaction between the user and the images in the volumetric display by using a concave mirror to project the images in a volumetric display the user's space, and by using a method of virtual manipulating device to extend user's motion into the volumetric display space."); **12:49-55**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called"volumetric image projector"").

Element 3:

using one or more sensors coupled to a computer processor,

Tsao discloses one or more sensors coupled to a computer processor (For e.g., sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with system. According to this prior art, sensor such as the position tracker is connected to a computer.).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 4:

sensing a user interaction with the displayed initial image,

Tsao discloses sensing a user interaction with the displayed initial image (For e.g., sensors , such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images/ initial image which is possible only because of sensors.).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."). Fig.23

Element 5:

the user interaction being in a three dimensional coordinate system,

Tsao discloses the user interaction being in a three dimensional coordinate system (For e.g., "reach into" the 3D data image to perform desired data handling or manipulation is the user interaction in three dimensional coordinate system.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 6:

and said sensing being based on quadrilateral angle navigation

Tsao discloses the sensing being based on quadrilateral angle navigation (For e.g., sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system or configured based on quadrilateral angle navigation. Position tracking system can be based on quadrilateral navigation).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 7:

that determines a sensing point position within the three dimensional coordinate system,

Tsao discloses sensing that determines a sensing point within the three dimensional coordinate system (For e.g., sensing by sensors such as position tracking system, 3D coordinate measurement arm can be used to determine a sensing point position within three dimensional coordinate system similar to quadrilateral angle navigation).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation.");**12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 8:

wherein an original image is displayed from a certain point in space

Tsao discloses an original displayed from a certain point in space (For e.g., 3D images (including original image) are projected which seem floating in mid air 2350 near the optical axis of the concave reflector or in other words, it can be said that the original image is displayed from a certain point in space.).

*See e.g.,***12:49-55**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector".").

Element 9:

preconfigured to match specific points in the quadrilateral angle navigation,

Tsao discloses an original image is displayed from a certain point in space preconfigured to match specific points in the quadrilateral angle navigation (For e.g., 3D images (including original image) are projected which seem floating in mid air 2350 near the optical axis of the concave reflector or in other words, it can be said that system is configured to match specific points in mid air 2350 near the optical axis of the concave reflector. Position tracking is based on quadrilateral angle navigation thus, said matched specific points are in quadrilateral angle navigation.).

*See e.g.,***12:49-58**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector,as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 10:

and the certain point in space is matched to space location of the user interaction to determine the sensing point,

Tsao discloses that the certain point in space is matched to space location of the user interaction to determinse the sensing point (For e.g., 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface allow user to interact directly with the 3D image. Sensor such as position tracking device tracks the user interaction in 3D to determine the sensing point.).

*See e.g.,***12:49-58**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Fig.23

Element 11:

said sensing resulting in each sensor providing coordinate information to the computer processor

Tsao discloses sensing resulting in each sensor providing coordinate information to computer processor (For e.g., Sensing is done in 3D therefore position information which includes coordinates is provided to connected computer for processing.).

*See e.g.,***12:55-58**("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 12:

enabling the computer processor to correlate with the original image in the three dimensional coordinate system;

Tsao discloses enabling the computer processor to correlate with the original image in the three dimensional coordinate system (For e.g., user interaction data is sensed and provided to the computer to process it. For user is able to manipulate and handle the image data, it is clear that computer correlates the sensed data with initial (original) image.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 13:

in the computer processor, correlating the sensed user interaction with the three dimensional coordinate system

Tsao discloses that in the computer processor correlating the sensed user interaction with the three dimensional coordinate system (For e.g., user interaction in three dimensional system is sensed by sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface. This data is provided to the computer to process/ correlate in three dimensional system.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display;the position tracker being connected to said computer;").

Element 14:

and responsively

Tsao discloses that in the computer processor action is taken in response to user interaction sensed data (For e.g., user is able to interact with the 3D image directly which can be seen in fig. 23. Thus, it can be said that to directly interact, system should be responsive to interaction sensed data.).

See e.g. **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display;the position tracker being connected to said computer;").

Element 15:

manipulating the initial image

Tsao discloses manipulating the initial image in response to user interaction (For e.g., this prior art provides facility to manipulate the initial 3D image data by user which can be reflected after processing the interaction sensed data by computer).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation.").

Element 16:

resulting in an updated image

Tsao discloses correlating the sensed user interaction with the three dimensional coordinate system and responsively manipulating the initial image resulting in an updated image based on the correlated user interaction (For e.g., this prior art discloses an invention in which user is able to handle the 3D image data and manipulate it directly. Manipulation of image is visible after user interaction sensed data is processed/ correlated by the computer connected with the sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display;the position tracker being connected to said computer;").

Element 17:

based on the correlated user interaction;

Tsao discloses an updated image based on the correlated user interaction (For e.g., this prior art discloses an invention in which user is able to handle the 3D image data and manipulate it directly. Manipulation of image is visible after user interaction sensed data is processed/ correlated by the computer connected with the sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 18:

and from the computer processor through a projector,

Tsao discloses projecting method of the updated image which involves the processing from the computer through a projector (For e.g., computer connected with the sensors processes/ correlates the data and updated image is displayed using "volumetric image projector").

*See e.g.,***12:41-45**("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:49-58**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 19:

projecting the resulting updated image,

Tsao discloses projecting of the updated image (For e.g., user is able to handle the 3D image data or manipulate it directly. This sensed data is sent to the computer connected with the sensors which processes/ correlates the data and updated image is displayed using "volumetric image projector").

*See e.g.,***12:41-45**("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:49-58** ("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector".A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

Element 20:

the updated image being equivalent to the initial image

Tsao discloses the updated image which is the updated version of the original image before interaction (For e.g., user is able to manipulate the original image directly. In fig. 24, user "picks up" a tissue surface 2490 and that is reflected in the updated image. Concept of fig. 24 is can be applied on "volumetric image projector" of fig. 23.).

*See e.g.,***12:49-58** ("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.").

*See e.g.,***13:36-49** ("Haptic interface such as force-feedback systems can be integrated with this system to create "virtual tools". For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The "virtual end" 2471 can be defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to "pick up" a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The "virtual manipulator" concept can, of course, also be used in the "volumetric image projector".").

Element 21:

manipulated by a distortion in three dimensional space.

Tsao discloses that the updated image is projected which reflects the distortion by manipulation in original image (For e.g., user is able to manipulate the original image directly. In fig. 24, user "picks up" a tissue surface 2490 and that is reflected in the updated image. Concept of fig. 24 is can be applied on "volumetric image projector" of fig. 23.).

*See e.g.,***12:49-58** ("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the systemto allow the user **2370** interact directly with the 3D images.").

*See e.g.,***13:36-49** ("Haptic interface such as force-feedback systems can be integrated with this system to create "virtual tools". For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The "virtual end" 2471 can be defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to "pick up" a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The "virtual manipulator" concept can, of course, also be used in the "volumetric image projector".").

* * * * *

End of Report

Please rank my report. The following link leads to our profile at ClaimCharts.info, where you could rank our report.
Rank Here

Evaluation Code:

We are available to answer follow up questions, provide clarifications and search for additional materials upon request. For public contact - please refer to our profile at ClaimCharts.info. For private contact - please call/email us.

Our hourly rate is: $

Non-Disclosure Agreement:

We understand that for the purpose of assisting you in collecting data, you may disclose to us certain confidential information that you want to keep confidential; therefore, for good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, we agree as follows:

1. For a period of twelve (36) months from the date we receive your confidential information ("Effective Date"), we will not use or disclose the information, except: (a) in the event that such information (i) comes into the public domain not through us; (ii) is already known to us at the time of disclosure; (iii) is lawfully otherwise learnt or independently developed by us; or (iv) is required to be disclosed by law.

2. Since a breach of our obligations hereunder may result in irreparable harm to you, we agree that upon such breach or threat thereof, you shall be entitled to specific performance and other injunctive relief without posting a bond, and in the event of disclosures required by court order, we shall promptly notify you of the proceeding.

3. This agreement supersedes all prior discussions and writings and constitutes the entire agreement between you and us with respect to the subject matter hereof. This Agreement shall be binding upon the successors and assigns of the respective parties, and may not be modified except by an instrument executed by both parties.

Saurabh singh